

Chapter 1: Optical Network Unit Specifications

1.1 Overview

The ONU (Optical Network Unit) is a key component in GPON networks.

1.2 Receiver Parameters

Parameter	Value	Unit
Rx Sensitivity	-28	dBm
Rx Wavelength	1490	nm
Rx Overload	-8	dBm

1.3 Transmitter Parameters

Parameter	Value	Unit
Tx Power	+3	dBm
Tx Wavelength	1310	nm
Tx Extinction Ratio	10	dB

1.4 Relationship

The ONU connects to the OLT (Optical Line Terminal) via optical fiber.
The OLT manages multiple ONUs through the GPON protocol (ITU-T G.984).

Chapter 2: Network Topology

2.1 Basic Topology

OLT --[Optical Fiber]--> Splitter --[Distribution Fiber]--> ONU_1 --> ONU_2 --> ONU_3

Chapter 3: EPON Specifications

3.1 EPON Parameters

The EPON upstream wavelength is 1310nm.
The EPON downstream wavelength is 1490nm.
Rx Sensitivity for EPON ONU is -27dBm.